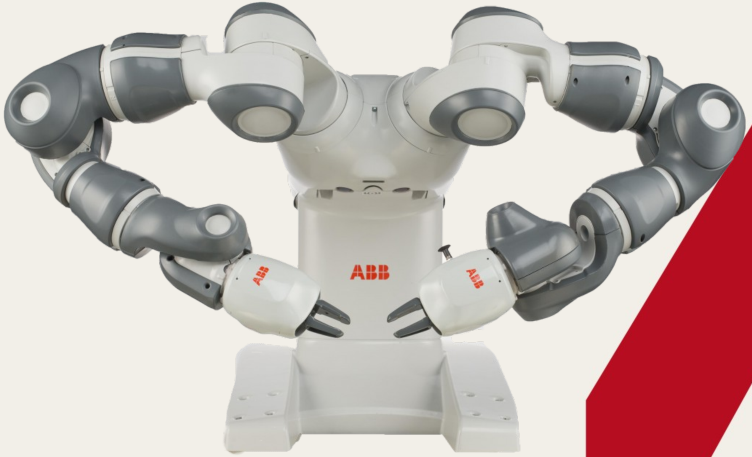
Robot

A robot is a machine especially one programmable by a computer capable of carrying out a complex series of actions automatically. [1]

[2, 4, 5]

Industrial Robot

Industrial robots are considered as a cornerstone of competitive manufacturing, which **aims to combine high productivity, quality, and adaptability at minimal cost**.



[12, 16, 17]

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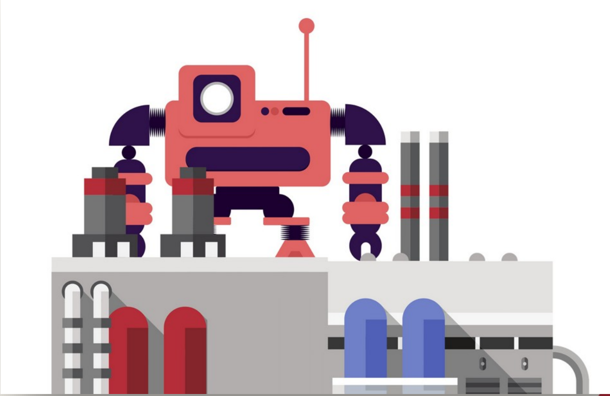
Automation

Automation describes a wide range of technologies that reduce human intervention in processes. Human intervention is reduced by predetermining decision criteria, subprocess relationships, and related actions and embodying those predeterminations in machines. [1]

- Discrete event control
- Programmable Logic Controller
- Sequential control
- PID
- Supervisory control
- Data Acquisition
- Distributed control system
- SCADA
- Human Machine Interface
- Sensor
- Actuator



https://www.pxhere.com

Industrial Robot Types

Parallel (Parallel link configuration)

Advantages
Faster
Higher acceleration
Better load distribution
Greater accuracy
Greater rigidity
Higher energy efficiency
High stiffness at low inertia

Disadvantages
Relatively expensive
Limited workspace
Small payload
Requires large framing and space

Serial (Rigid links connected with revolute joints)

Advantages:
Easier maintenance
Variable payload
Low cost
Durable
Large workspace

Disadvantages:
Relatively slow
Power consumption
Less accurate
Accumulative error

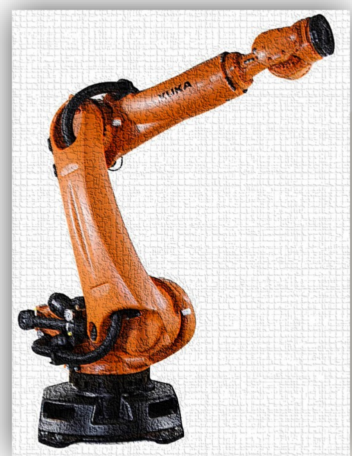
[16]

Robot Manufacturers

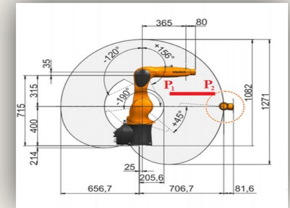
- ABB
- KUKA
- NACHi
- COMAU
- FANUC
- MITSUBISHI
- YASKAWA
- Omron

Industrial Robots: Selection Criteria

DOF



Application



Workspace

Payload

Accuracy

Speed

£

Price

Energy consumption [12, 18]

Industrial Robots: Selection Criteria

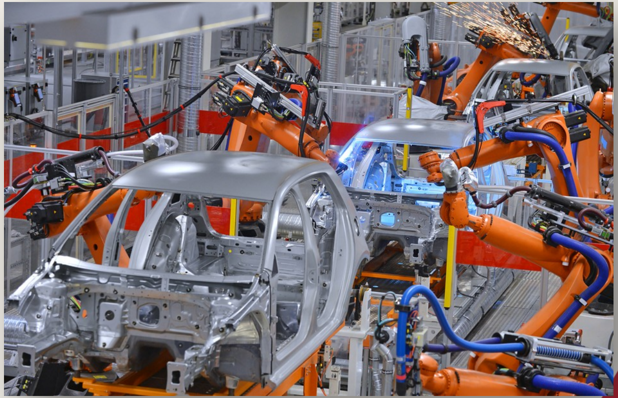
[12]

Serial Robot

Six Degree of Freedom (DOF) or more (redundant configuration), Payload: 500g to 1000 Kg.

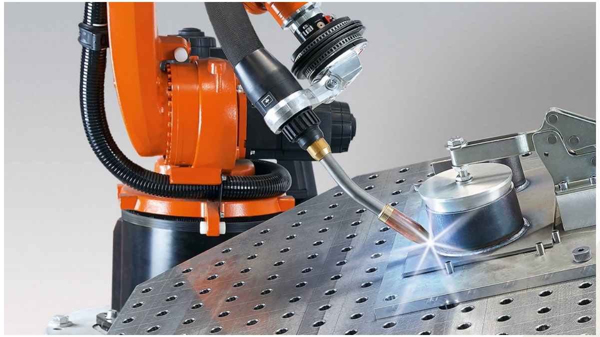
Applications

- Welding and Grinding (Blending, Weld grinding)
- Assembly task
- Painting
- Material Transfer
- Machining
- Human–Robot Cooperation
- Die Casting
- Injection Molding
- Laser welding and cutting
- Dispensing
- Inspection and measuring
- Palletizing and packing



Robot welding

- High repeatability and positioning accuracy ≤ ±0.1 mm.
- High speeds of the end-effector of up to 8 m/s.
- Larger reachability than human
- Higher quality
- Lower cost
- Force calibration

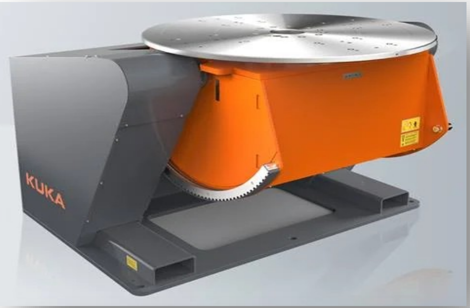


Arc welding

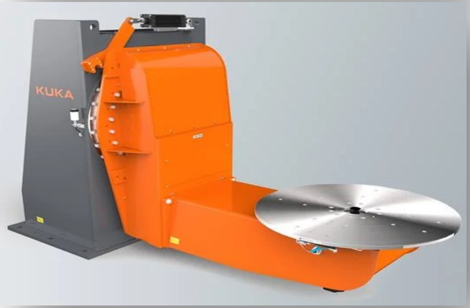
Preconfigured spot welding automation package

[12]

Robot Peripheral





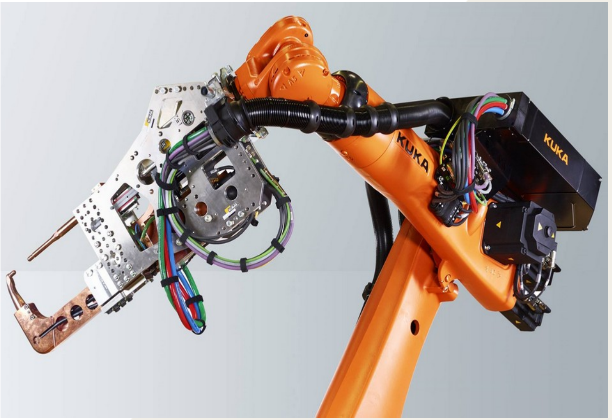



Positioners and Turntables

[12]

Robot Peripheral

Preconfigured arc welding accessory

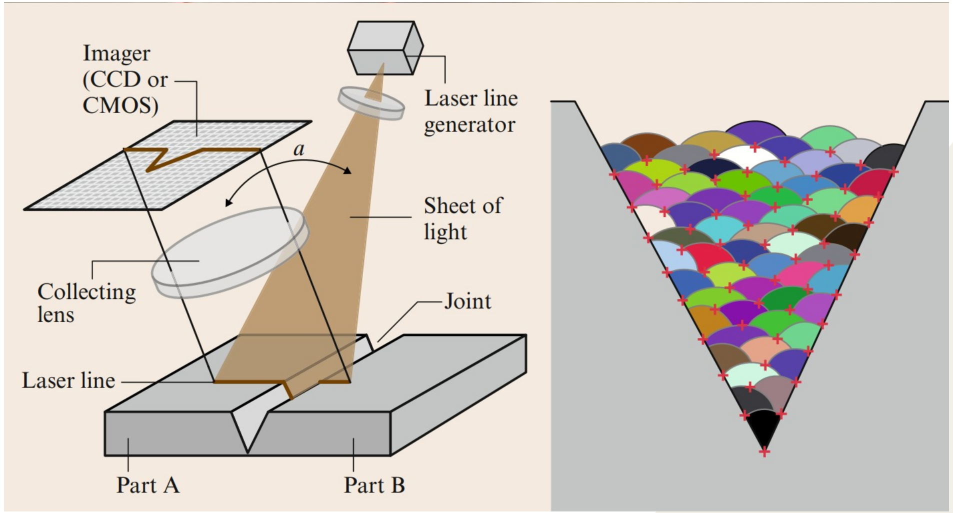


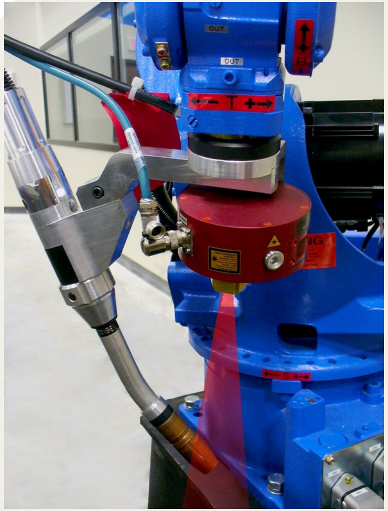
Spot welding Servo gun+fume extractor

Floor-mounted linear unit

[12]

Seam Sensor



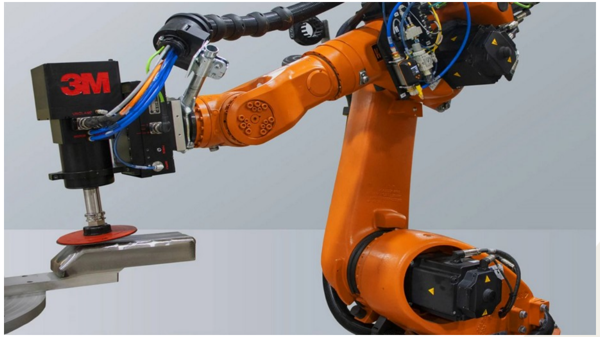


A seam sensor is based on the laser triangulation principle. The sensor is attached to the robot flange. A laser projects a sharp stripe on the seam. The stripe is detected by a 2D CCD sensor. From the extracted contour the position of the seam relative to the sensor is calculated.

[7, 13]

Robot Grinding

- Higher productivity
- Faster
- Safer
- Higher quality grinding and consistent surface
- Lower cost



[12]

Robot Painting

- Safer
- Faster
- Optimal throughput and accessibility using multiple robots

- Using simulation software the painting process can be optimised
  - Optimise paint deposition
  - Thickness
  - Coverage

- All paint materials can be applied including
  - solvent-based paints
  - water-borne paints
  - powder

[12]

Selective Compliance Assembly Robot Arm (SCARA)

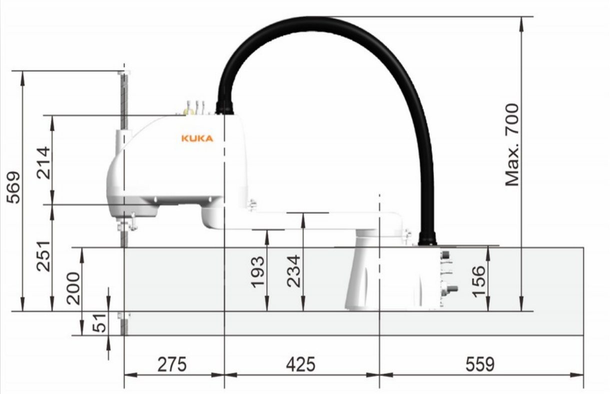
- Suited for assembly tasks
- Low-cost
- Rigid in the vertical axis
- Compliance in the horizontal axis
- Three or four rotational and one translational axis
- Relatively smaller workspace
- Limited motion in vertical axis
- Limited payload (less than 7Kg)

[16]

SCARA Robot

SCARA family is designed for a variety of general purpose applications. Such as:

- Tray kitting
- Component placement
- Machine loading/unloading
- Assembly (small size and weight)
- Laboratory automation
- Drug dispensing



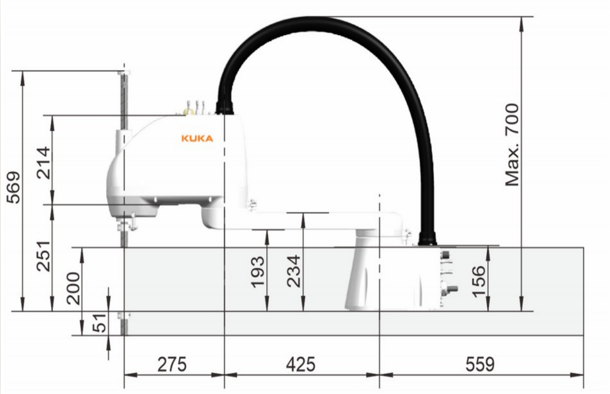
These applications require fast, repeatable and articulate point-to-point movements such as palletising, depalletising, machine loading/unloading and assembly.

[12]

SCARA Robot

Advantages

- Fast (± 720°/s)
- Repeatable Pose repeatability ± 0.02 mm
- Rapid cycle times (up to 200 cycles per minute)
- High precision and reliability.
- Low cost
- Easy and low cost maintenance
- Table top mountable
- Modular design



Disadvantages

- Low payload
- Small workspace
- Limited tasks/applications

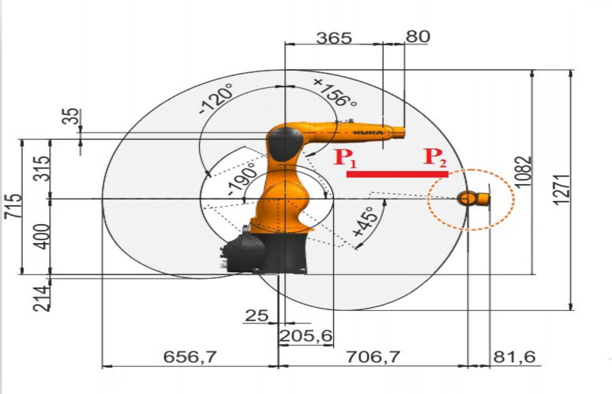
www. global.abb/group/en

[12]

Serial Robot

Advantages:
- Most common industrial robots
- Easier maintenance
- Variable payload
- Low cost
- Durable
- Large workspace
- Multi purpose applications
- Smaller number of singularities
- Require larger space

Disadvantages:
- Relatively slow
- Higher Power consumption
- Relatively smaller accuracy and repeatability
- Smaller rigidity
- Heavy
- Costly maintenance



[12]

Serial Robot Mounting

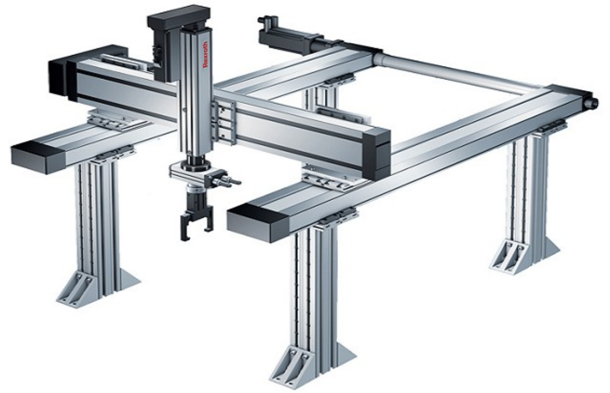
[12, 16]

Cartesian Robot

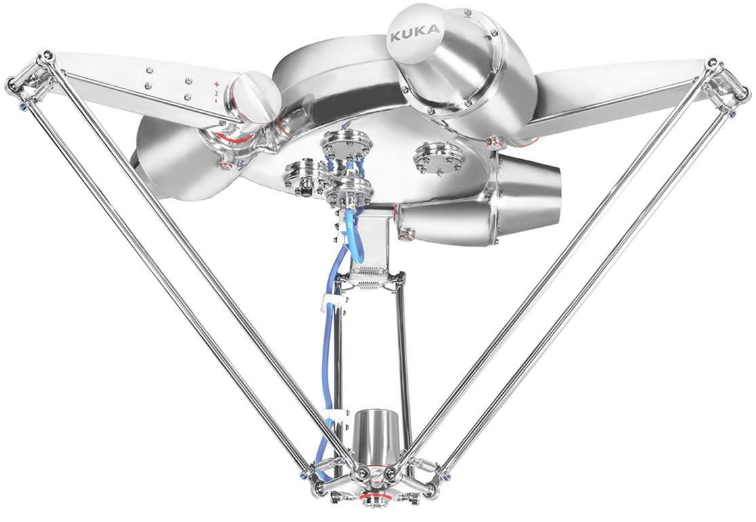
- Ideally suited for covering large workspaces.
- Three orthogonal translational axes (three prismatic joints)
- Three DOF (XYZ)
- Larger workspace
- High speed and precision.
- No singularity
- Simple inverse kinematic
- Flexible design.
- Electric, Pneumatic or hydraulic actuator.

Application

- Particularly valuable in logistics, 3D printers, CNC



Parallel Robot



Delta robot three revolute joints (servo motors)

Stewart six Prismatic joints

[11, 12]

Parallel Robot

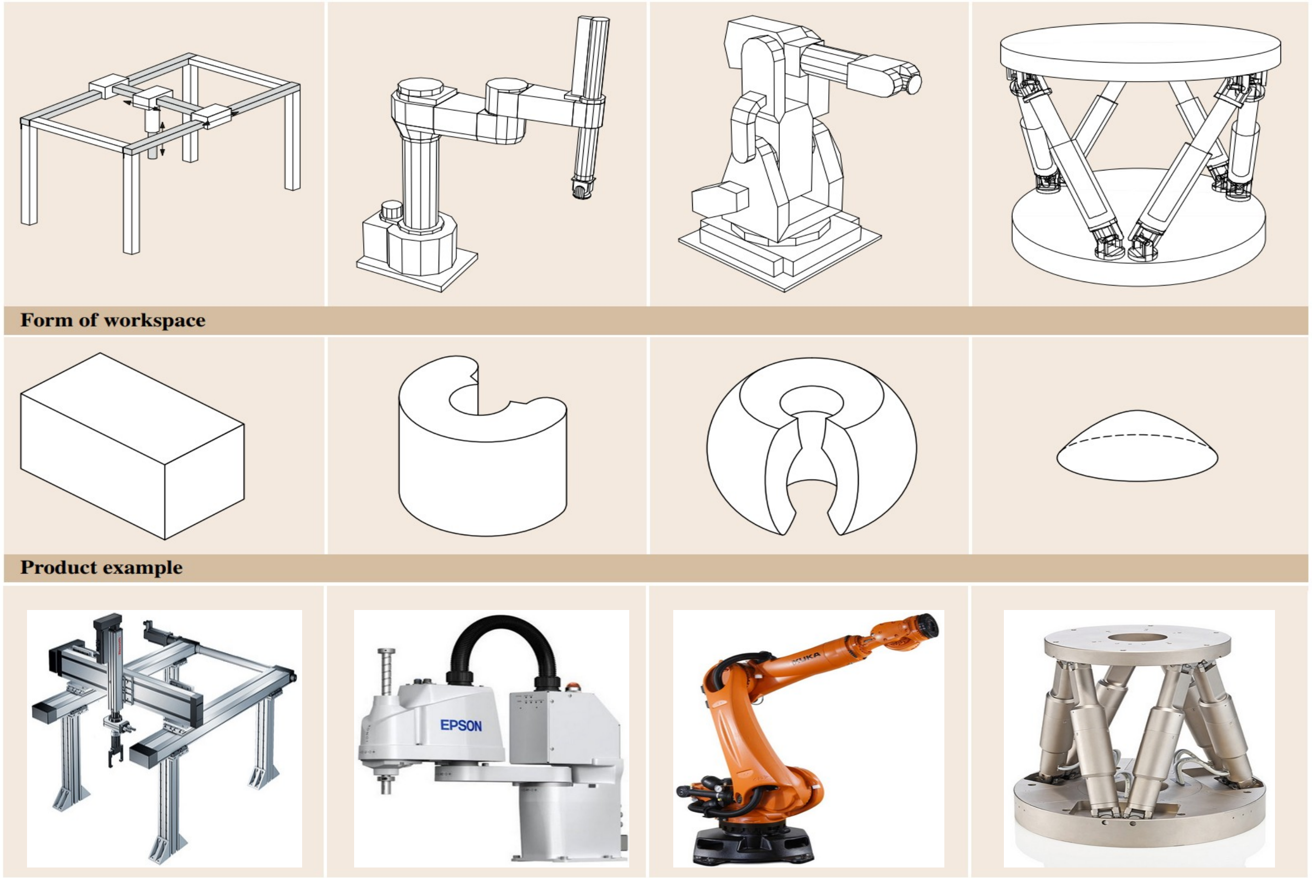
Advantages

- Higher accuracy due to the averaged position error
- High stiffness
- Low inertia
- Power efficient
- Accurate, high speed motions
- Shortest cycle times
- Payloads from 1 to 8kg

Applications

- Food industry
- packing application
- handling of flow wrapped products, Meat, fish, vegetables.
- sorting and packaging applications
- Food industry, bread, biscuit …

[16]



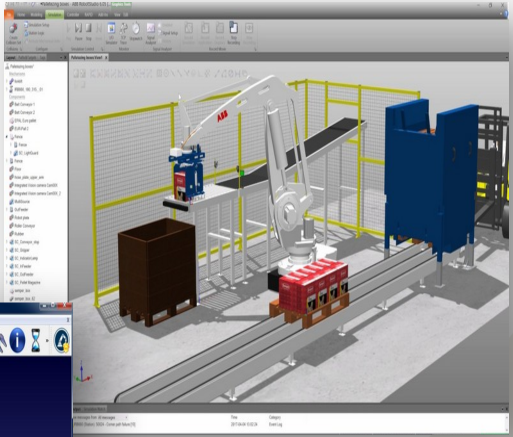
[13]

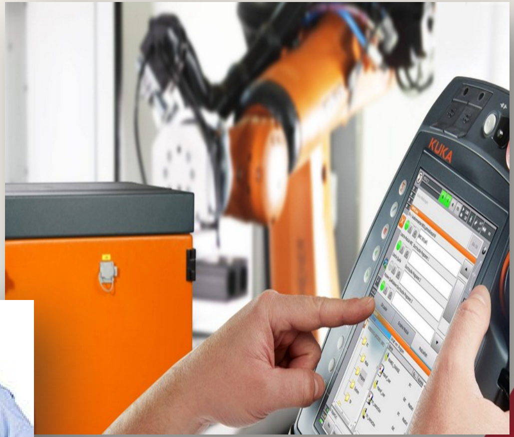
Collaborative Robot (Cobot)

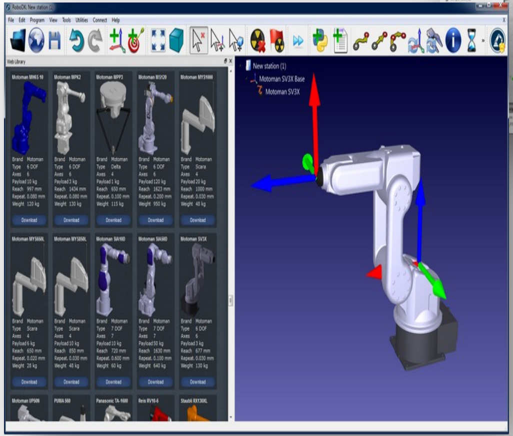
Robots with Impedance control, force feedback, FT sensor and back driveable compliant joints. Designed for human robot interaction.

https://industrial.omron.eu/en/ https://www.universal-robots.com

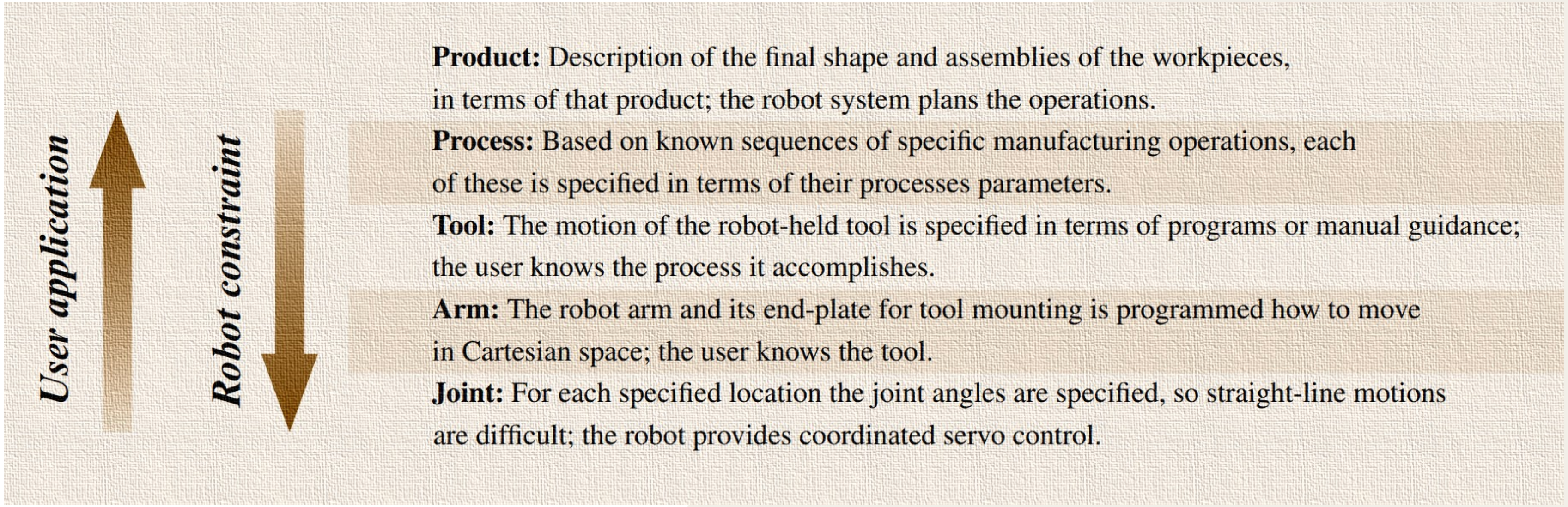
Robot Programming







https://www.phase1vision.com/



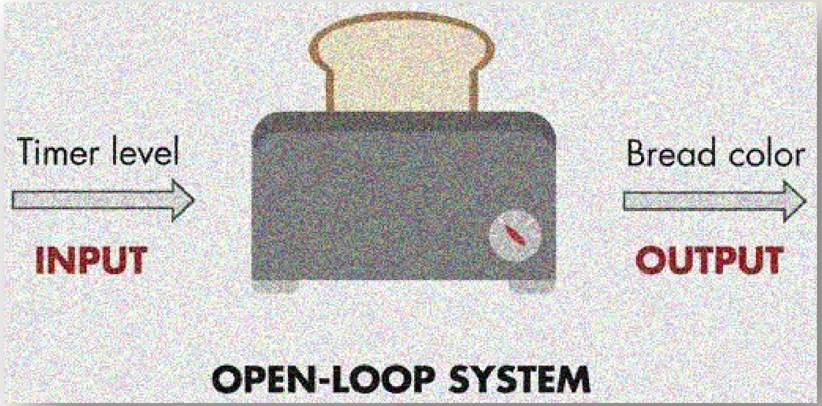
Robot-related centricity, ranging from a high-level view of the work to be accomplished in terms of the product to manufacture, to a low-level robot motion view that, in practice, constraints what manufacturing operations can be performed.

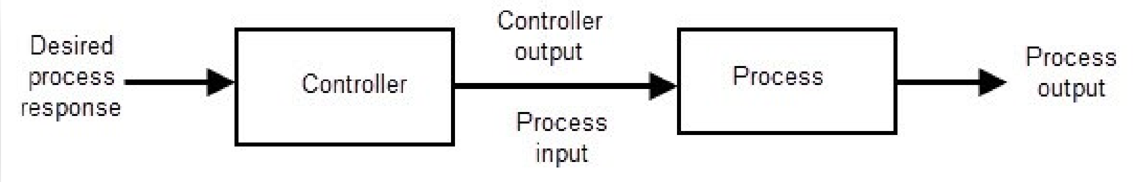
[13]

Control System

A control system monitor, manages, commands, control, or regulates the behaviour (output) of other devices or systems using feedback loops. It can range from a simple home temperature controller using a thermostat or a domestic boiler control to large scale industrial control systems which are used for controlling processes or machines. [1]

Open-loop control system





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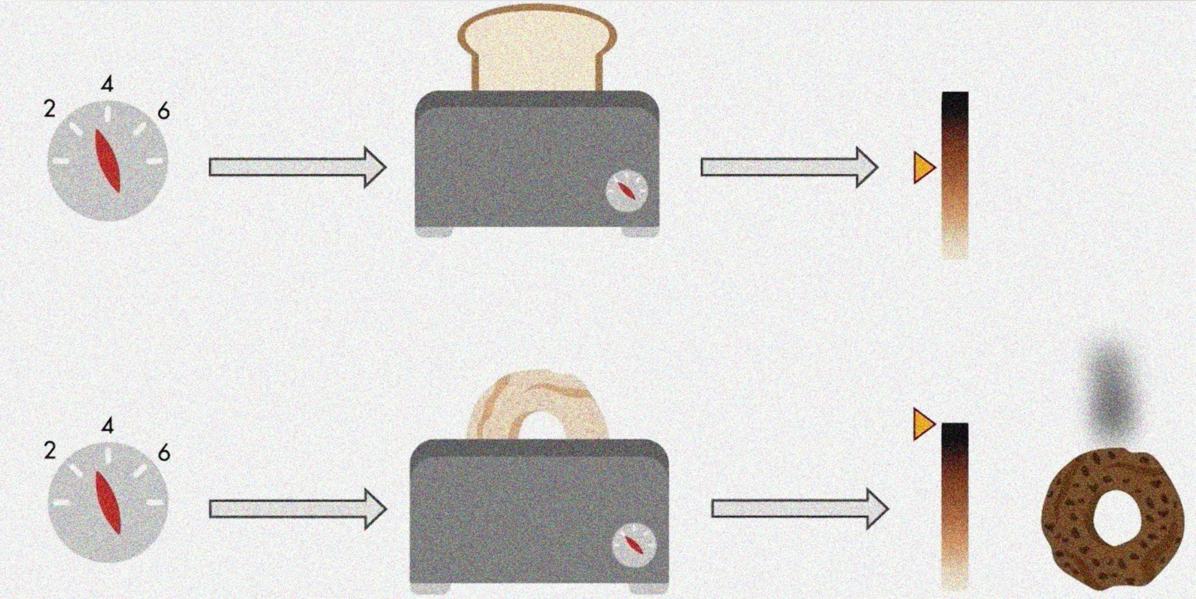
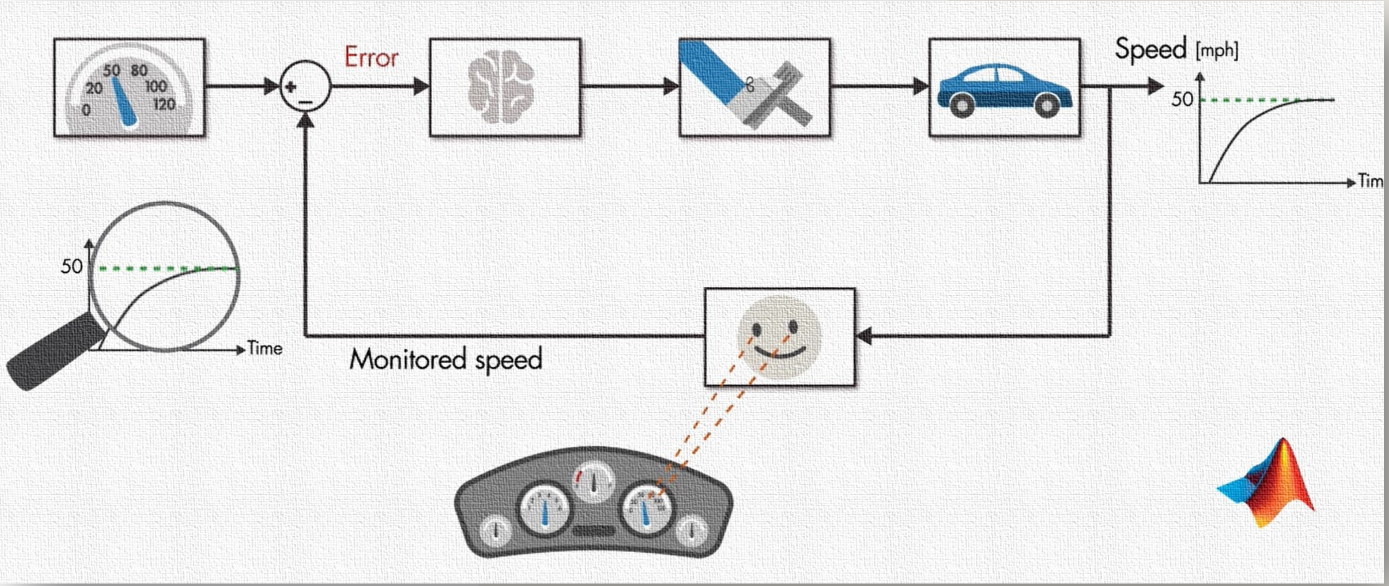
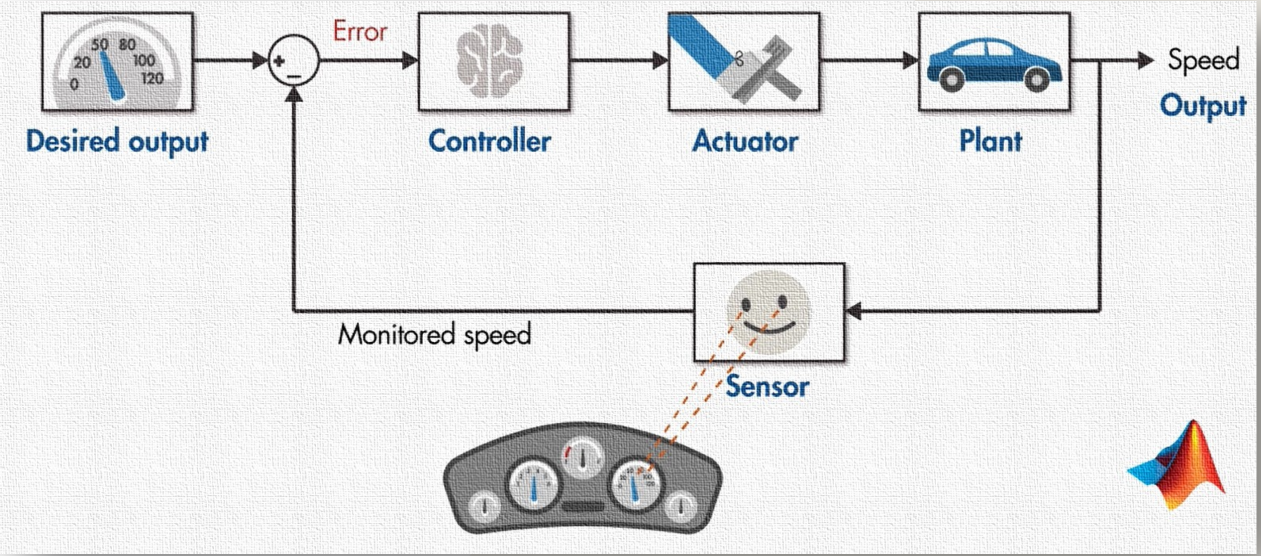
Open-loop control system



Closed-loop control system



Closed-loop control system



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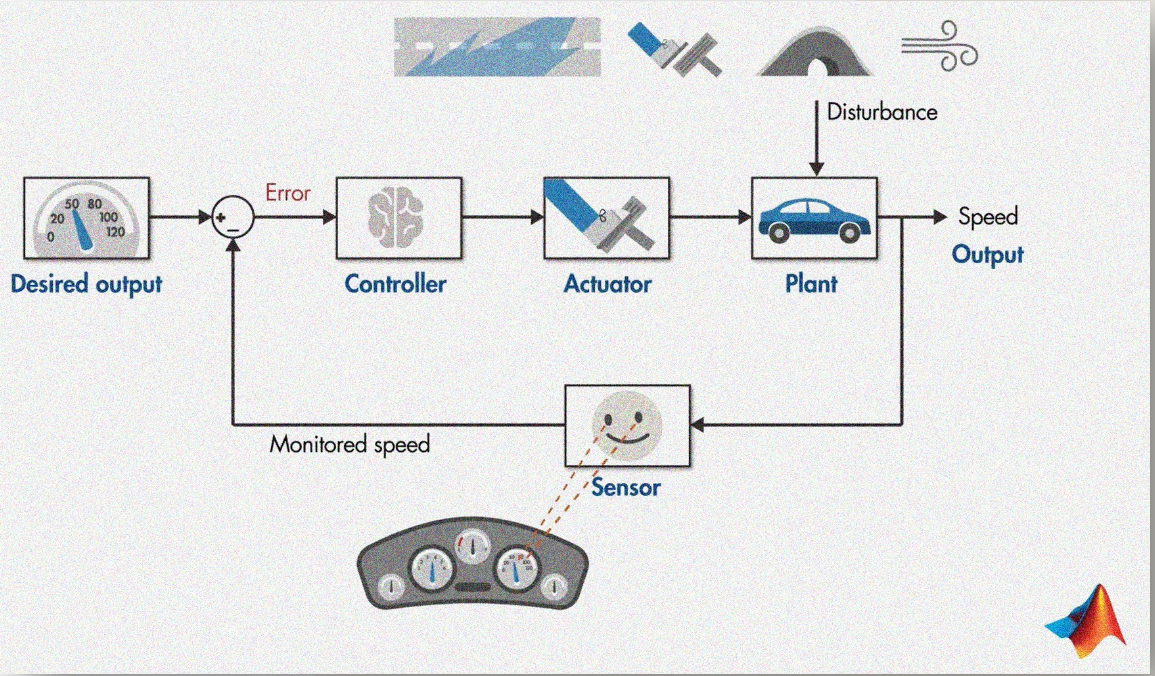
Closed-loop control system



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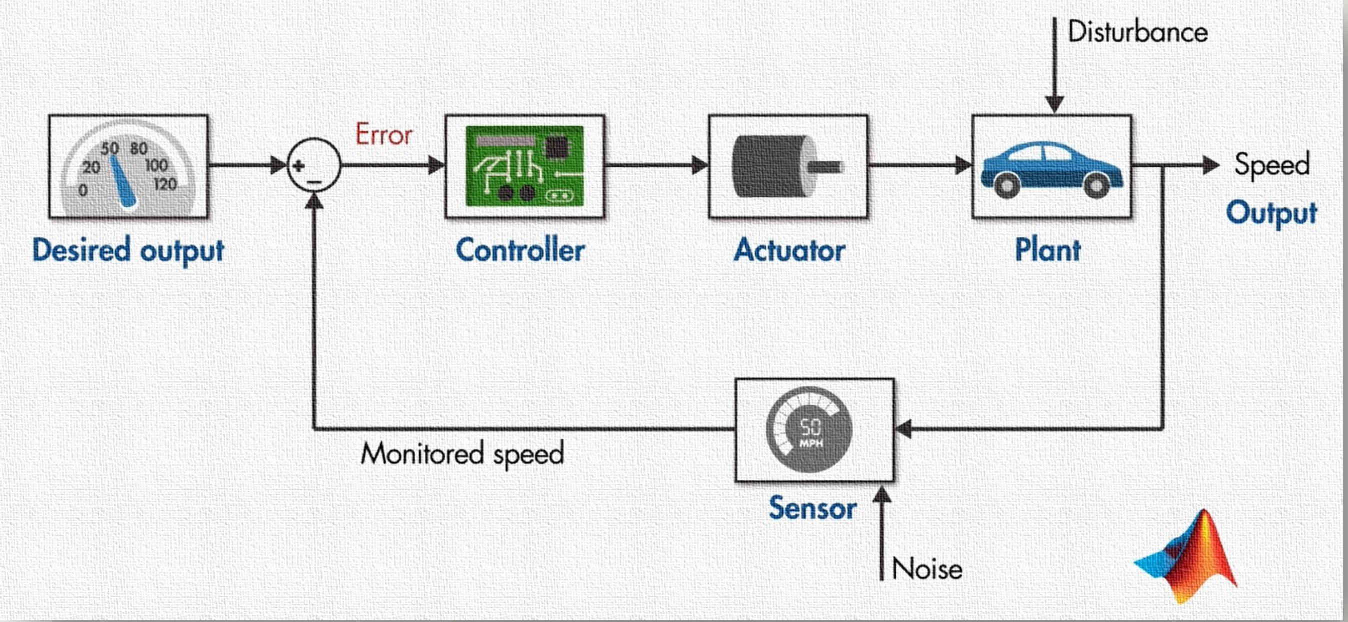
Closed-loop control system



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Closed-loop control system limitation

Plant/process limitation

Closed-loop control system

The **PID** controller is the most common form of feedback control. It was an essential element of early governors and it became the standard tool when process control emerged in the 1940s.

In process control today, more than 95% of the control loops are of PID type, most loops are actually PI control. PID controllers are today found in all areas where control is used.

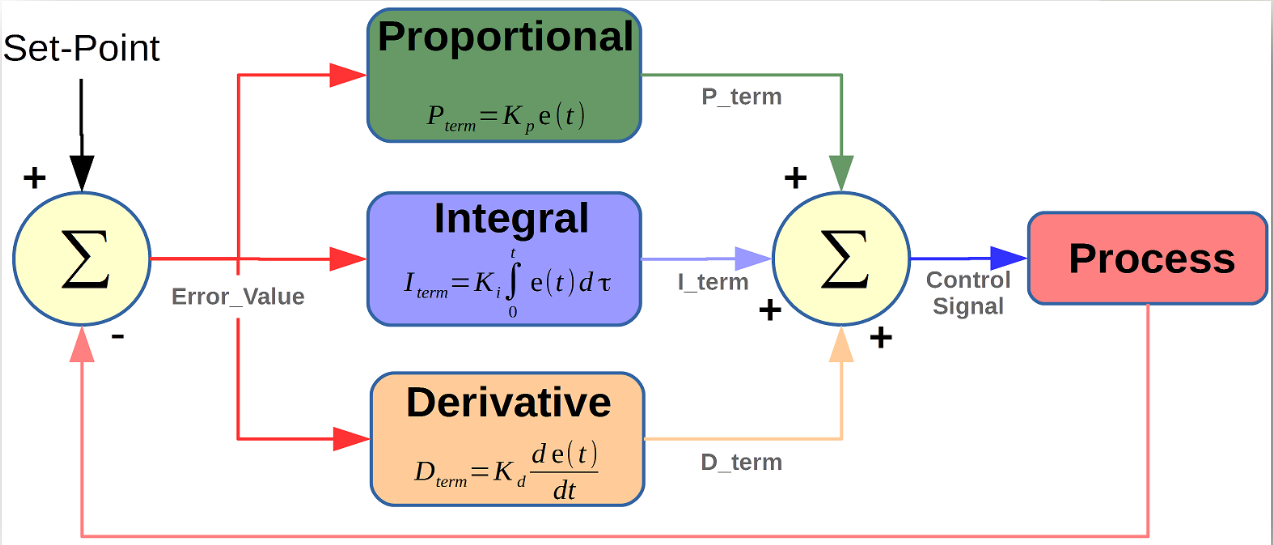
PID controllers come in many different forms. There are standalone systems in boxes for one or a few loops, which are manufactured yearly by the hundred thousands. PID control is also often embedded in many special purpose control systems, and is an important ingredient of a distributed control system.

Closed-loop control system

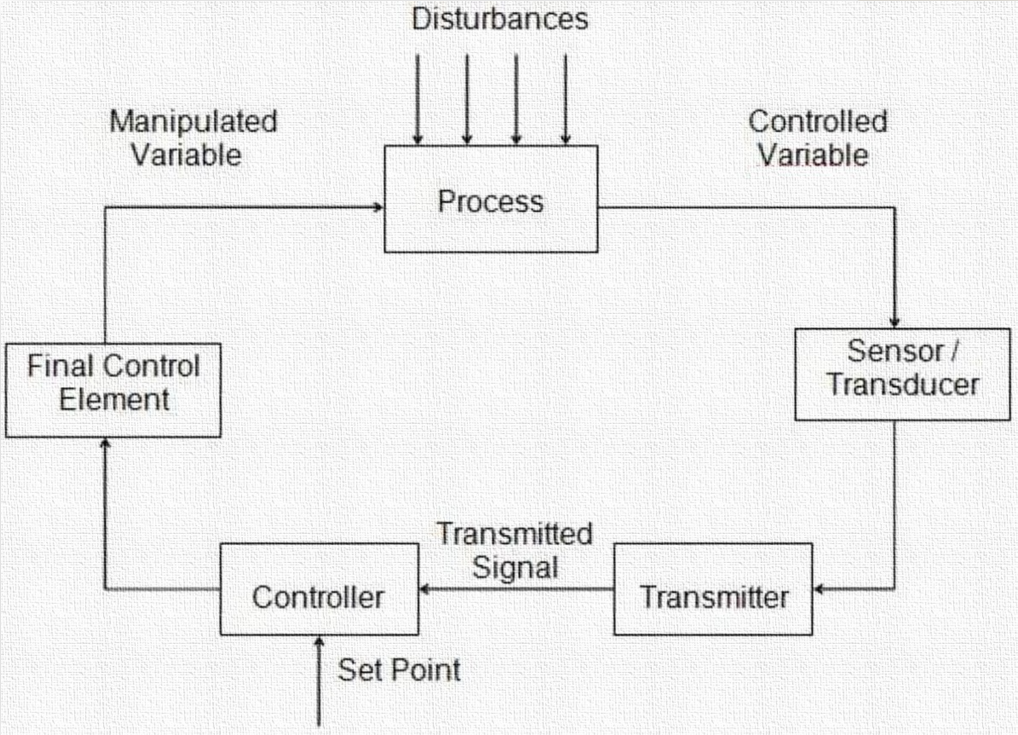
PID controllers have survived many changes in technology, from mechanics and pneumatics to microprocessors via electronic tubes, transistors, integrated circuits.

The microprocessor has had a dramatic influence on the PID controller. Practically all PID controllers made today are based on microprocessors. This has given opportunities to provide additional features like automatic tuning, gain scheduling, and continuous adaptation. In conclusion, the PID controller is an important component in every control engineer's toolbox.

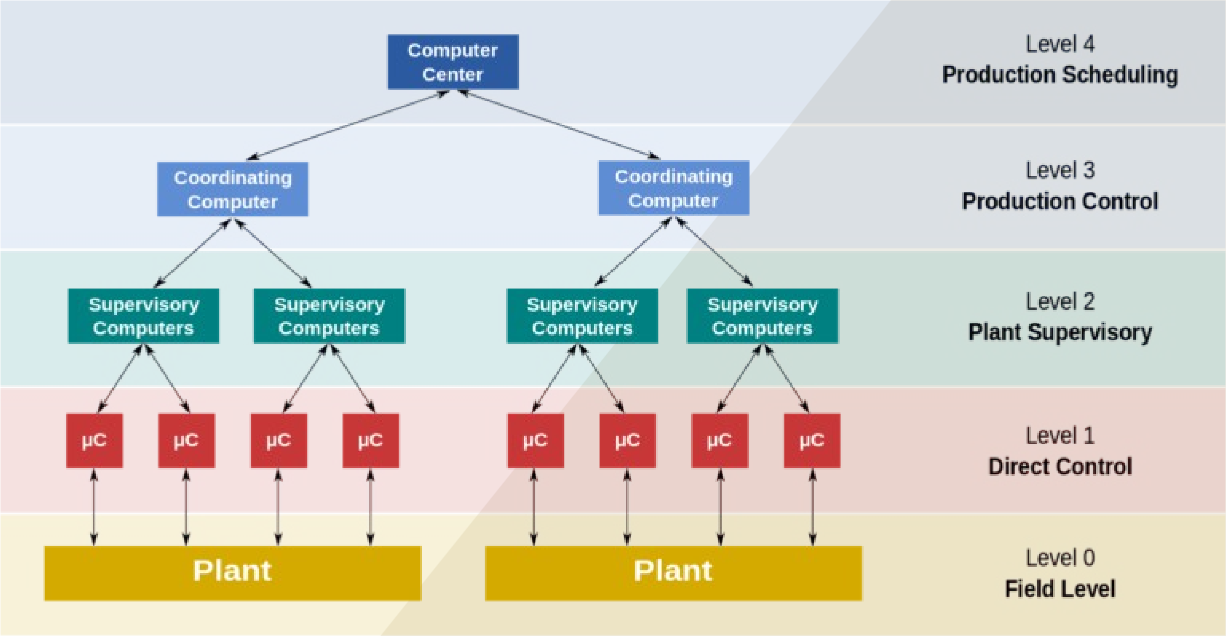
Closed-loop control system



Closed-loop control system



SCADA/DCS



Both SCADA/PLC and DCS types of industrial control system are based on the hierarchical structure shown in this figure.

- Level 0 contains the field devices such as flow and temperature sensors, and final control elements, such as control valves.

- Level 1 contains the programmable logic controllers (PLCs) or remote terminal units (RTUs).

- Level 2 contains the SCADA software and computing platform. The SCADA software exists only at this supervisory level as control actions are performed automatically by RTUs or PLCs.

- A feedback control loop is directly controlled by the RTU or PLC, but the SCADA software monitors the overall performance of the loop.

- Level 3 is the production control level, which does not directly control the process, but is concerned with monitoring production and targets.

- Level 4 is the production scheduling level.

Levels 3 and 4 are strictly not process control in the traditional sense, but are the levels where production control and scheduling takes place.

SIEMENS

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